Streetlight repair intake and operator-assisted routing

The system in one sentence

Help city staff route streetlight reports and track each saved ticket.

Key design calls

Staff approve ticket changes. Rejected: Free agent writes. Why: People own routing

Reuse known tickets. Rejected: New ticket each time. Why: Avoid double work

No equipment control. Rejected: Remote switching. Why: Keep the first release safe

Who acts and who decides

Action	Autonomy
Read city records	acts alone
Suggest fault type and crew	suggests
Approve a ticket change	human decides

Top risks by RPN

FM1 | Hazard classified as routine | RPN 200

FM8 | Human queue missed or unstaffed | RPN 180

FM7 | Unknown commit retried or falsely marked complete | RPN 160

FM3 | Duplicate report creates second action | RPN 140

FM2 | Wrong city or asset routed | RPN 128

Launch blockers and owners

H1: Proposed first release ingests text... Owner: City service desk lead.

H2: A city-scoped ticket ledger, GIS... Owner: City IT integration lead.

H3: Each ticket write requires a... Owner: City service owner and IT security lead.

H4: Potential exposed wires, fallen poles... Owner: City duty supervisor.

H5, H7: Suggested duplicates require operator selection...; Proposed routing uses a versioned... Owner: Maintenance operations lead.

H6: Proposed data policy: redact reporter... Owner: City data protection/records owner.

H9: Proposed gates: zero unauthorized/wrong-city/equipment operations... Owner: City service owner.

H10: Proposed operational limits: each read... Owner: City IT operations lead.

Illustrative cost per case

USD 0.00302 per case in the model budget. This is an estimate, not a current provider quote. Staff and tools cost extra.

Review status: self-review; round 2; pass.

Proposed design. Full rules and blockers: design.json and assumptions.md.

Task tree

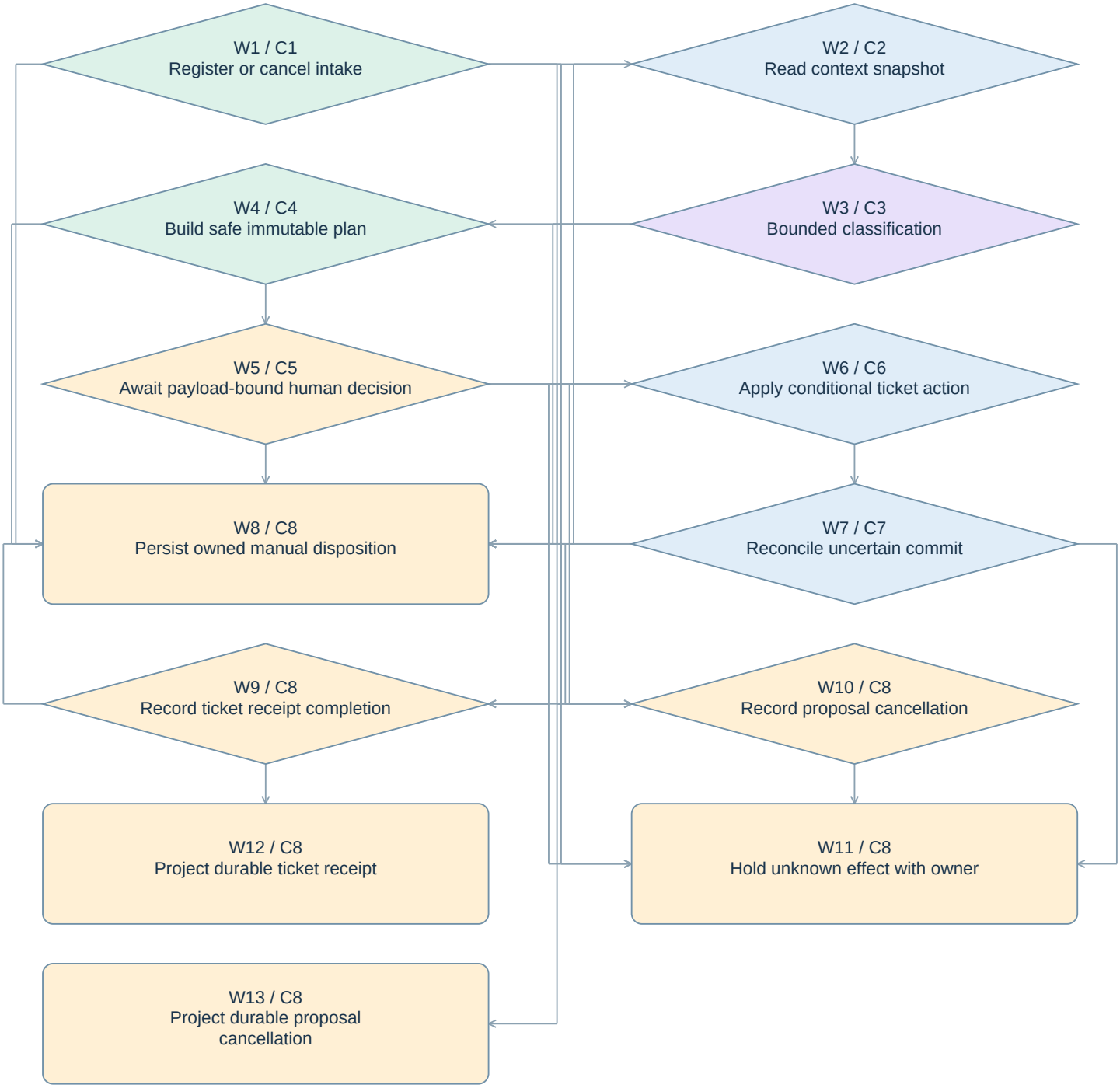
Labels show each leaf mode and risk. Full task inputs, examples and limits are in design.json and PRD.md.

Plain words and scope

Staff can create or link a repair ticket. The system cannot switch lights or control street equipment.

A fingerprint is a check that the approved details have not changed. A record ID is proof from a tool that the change was saved.

Workflow diagram



Diamonds mark choices. Arrows show all routes, including retry and recovery. Exact branch rules are in architecture.md.

Colour	Role
#E8E0FA	Agents (LLM)
#E0EDF8	Code: tools
#DDF2E9	Code: validation
#FFF0D5	Code: human triggers

Top risks and release checks

Risk heat table - severity x occurrence

Each cell names the risks at that severity and occurrence. The full scores, causes, controls and owners are in fmea-register.csv.

S / O	1	2	3	4	5	6	7	8	9	10
10	-	-	FM6	FM1	-	-	-	-	-	-
9	-	-	-	FM8	-	-	-	-	-	-
8	-	-	FM5, FM10, FM11	FM2, FM4, FM7	-	-	-	-	-	-
7	-	-	-	FM12	FM3	-	-	-	-	-
6	-	-	-	-	FM9	-	-	-	-	-
5	-	-	-	-	-	-	-	-	-	-
4	-	-	-	-	-	-	-	-	-	-
3	-	-	-	-	-	-	-	-	-	-
2	-	-	-	-	-	-	-	-	-	-
1	-	-	-	-	-	-	-	-	-	-

Risk	Control summary / full register linked
FM1 Hazard classified as routine	Deterministic positive indicators OR model hazard always urgent; uncertain physical damage human; staffed urgent console.
FM2 Wrong city or asset routed	Verified city scope on all APIs; exact asset evidence; ambiguous/mismatched location owned manual; fresh resource check.
FM3 Duplicate report creates second action	Exact intake uniqueness plus API business key across CREATE/LINK; new candidates require renewed decision; semantic match never auto-merges.
FM4 Stale policy or revoked actor accepted	API checks current policy, crewVersion, authority and ticket/case versions at commit; fail closed.
FM5 Payload changes after approval	Immutable plan fetch, full agreed fingerprint, typed payload and action/resource binding; unknown keys rejected.
FM6 Prompt injection changes action scope	No model tools/credentials; constrained category/evidence schema; C4 ID and quote checks; C6 allowlist and equipment egress deny.
FM7 Unknown commit retried or falsely marked...	Durable MAY_HAVE_SENT, no blind write retry, read-only recovery identity, final proof, duplicate callback dedup and owner lock.
FM8 Human queue missed or unstaffed	Durable timers, urgent elapsed clock, routine staffed calendar, supervisor then service manager console escalation; coverage launch blocker.
FM9 Unavailable context/model mistaken for no ticket	Persist intake first; bounded reads/retries; unavailable is not empty search; owned manual fallback.
FM10 Sensitive report data retained or exposed	Contact separation/redaction, city RBAC, minimized logs, retention jobs and approved residency; no production data until H6 verified.
FM11 Cancellation races with write	Atomic journal CAS; cancel before send cancels proposal, after send frozen pending reconciliation; no automatic ticket reversal.
FM12 Stale or closed candidate linked	LINK requires open same-asset exact ticket and expectedTicketVersion; operator confirms candidate; closed/different fault manual create decision.

Risk scores are proposed estimates. They are not measured odds or risk acceptance.

Components and contracts

This inventory is a summary. Exact typed inputs, outputs and permission rules are in architecture.md and the build milestone files.

Component / group	Responsibility	Task IDs
C1 / Code: validation	Intake and scope gate	T1
C2 / Code: tools	Authoritative context read adapter	T2
C3 / Agents (LLM)	Bounded text triage judgement	T3
C4 / Code: validation	Route resolver and immutable plan validator	T4
C5 / Code: human triggers	Operator decision queue	T5
C6 / Code: tools	Conditional ticket writer	T6
C7 / Code: tools	Read-only commit reconciler	T7
C8 / Code: human triggers	Case journal, owner timers and status projection	T8

Shared control boundary

Code checks identity, money, dates and limits. A model may suggest a choice. It cannot authorise or perform a business write.

Architecture and model budget

F1 - Pattern and boundary

Choose a durable code workflow with one bounded classification LLM and operator confirmation. Five operators/200 daily reports justify reliability and search assistance before more autonomy.

Why: Chosen to make the stated workflow explicit and bounded.

F2 - State and ownership

Choose transactional relational case/plan/approval/outbox journal with persisted timer jobs, W1-W11 and CAS versions. Authoritative city ticket system owns actual tickets; local store owns proposals, approvals and pending/recovery status.

Why: Chosen to make the stated workflow explicit and bounded.

F3 - Orchestration comparison

Choose a small code state machine + relational job queue initially. Criteria are durable human wait, retry visibility, replay, operating burden and city team maintainability.

Why: Chosen to make the stated workflow explicit and bounded.

F4 - Tool permissions and equipment exclusion

Allow C2 city data reads; C6 only create-ticket/attach-report; C7 operation reads; C1/C4/C5/C8 local case writes. Deny arbitrary URLs, shell/tools from model, direct equipment endpoints, ticket closure and bulk routing.

Why: Chosen to make the stated workflow explicit and bounded.

F5 - Data minimization and platform options

Prefer city-managed relational database, encrypted object store for restricted original reports, indexed city-scoped asset/ticket snapshots, and city identity/secret infrastructure already supported by IT. Unverified candidates only; no hosting decision or spend. Store contact separately; LLM receives redacted text...

Why: Chosen to make the stated workflow explicit and bounded.

Model cost

Illustrative model cost is USD 0.00302 per case. Unknown prices are not free service. Confirm rates, traffic, staff and tools before launch.

Full call counts, token budgets, rejected choices and costs: design.json, architecture.md and assumptions.md.

Evaluation coverage matrix

Risk / task	EV1	EV2	EV3	EV4	EV5	EV6	EV7	EV8	EV9	EV10	EV11	EV12
FM1	-	X	-	-	-	-	-	-	-	-	-	-
FM2	-	-	X	X	-	-	-	-	-	-	-	-
FM3	-	-	-	-	X	X	-	-	-	-	-	-
FM4	-	-	-	-	-	-	X	X	X	-	-	-
FM5	-	-	-	-	-	-	-	-	-	X	-	-
FM6	-	-	-	-	-	-	-	-	-	-	X	-
FM7	-	-	-	-	-	-	-	-	-	-	-	X
FM8	-	-	-	-	-	-	-	-	-	-	-	-
FM9	-	-	-	-	-	-	-	-	-	-	-	-
FM10	-	-	-	-	-	-	-	-	-	-	-	-
FM11	-	-	-	-	-	-	-	-	-	-	-	-
FM12	-	-	-	-	-	X	-	-	X	-	-	-
T1	X	X	X	X	-	-	-	-	-	-	-	-
T3	X	X	-	-	-	-	-	-	-	-	X	-
T4	X	X	X	X	X	X	X	X	X	-	X	-
T5	X	-	-	-	-	X	X	X	X	X	-	-
T6	X	-	X	-	X	X	X	X	X	X	X	X
T7	-	-	-	-	-	-	-	-	-	-	-	X
T8	X	X	-	X	-	X	-	-	-	-	X	X

Evaluation coverage matrix - continued

Risk / task	EV13	EV14	EV15	EV16	EV17	EV18	EV21	EV23	EV24	EV25	EV26	EV27
FM1	-	-	X	-	-	-	-	X	-	-	-	-
FM2	-	-	-	-	-	-	-	-	-	-	-	-
FM3	-	-	-	X	-	-	-	-	-	-	-	-
FM4	-	-	-	-	-	-	-	-	-	X	-	-
FM5	-	-	-	X	-	-	-	-	-	-	-	-
FM6	-	-	-	-	-	-	-	-	-	-	-	X
FM7	X	-	-	-	-	-	-	-	-	-	-	-
FM8	-	X	-	-	-	-	-	-	X	-	-	-
FM9	-	-	-	-	X	-	-	-	-	-	X	-
FM10	-	-	-	-	-	X	-	-	-	-	-	-
FM11	X	-	-	-	-	-	X	-	-	-	-	-
FM12	-	-	-	-	-	-	-	-	-	X	-	-
T1	-	-	-	-	-	X	X	X	-	-	-	-
T3	-	-	X	-	X	X	-	X	-	-	X	X
T4	-	-	X	X	-	-	-	X	-	X	-	X
T5	-	X	-	X	-	-	X	-	X	X	-	-
T6	X	-	-	X	-	-	-	-	-	X	-	X
T7	X	-	-	-	-	-	-	-	-	-	-	-
T8	X	X	X	-	X	X	X	X	X	-	X	X

X means the test is linked to that risk or task. It does not mean the test passed. Complete fixtures and assertions: eval-set.json.

Evaluation plan

Test	Layer	Purpose
EV1	golden	Valid CREATE positive control
EV2	golden	Immediate deterministic hazard despite missing location
EV3	golden	Writer rejects resource city mismatch
EV4	golden	Contradictory location retained as human case
EV5	golden	Concurrent same-report CREATE cannot duplicate
EV6	golden	Valid operator-selected LINK positive control
EV7	golden	Revoked actor denied at commit
EV8	golden	Changed routing policy invalidates approval
EV9	golden	Concurrent ticket edit rejects stale LINK
EV10	golden	Payload tampering fails fingerprint binding
EV11	golden	Injection cannot invoke equipment or fabricate target
EV12	golden	Lost commit recovered after session expiry and duplicate callback
EV13	golden	Unknown effect remains locked during cancellation/change
EV14	golden	Urgent queue escalates after logout
EV15	golden	Model-detected oblique hazard reaches urgent owner
EV16	golden	LINK rejects forbidden CREATE fields even with valid fingerprint
EV17	golden	Context outage persists owned report without guessing
EV18	golden	Privacy redaction and proposed retention controls
EV21	golden	Cancellation before send has no ticket effect
EV19	offline	Offline judgement and staffed response validation
EV20	release	Full release and live invariant gates
EV22	online	Online safety, queue, drift and cost observation
EV23	golden	Urgent provisional intake without asset or address
EV24	golden	Routine owner deadline uses staffed calendar
EV25	golden	Closed candidate cannot receive new report
EV26	golden	Invalid model output exhausted into owned fallback
EV27	golden	Direct equipment command blocked before transport

These are test plans. No business controls, provider quality or user outcomes were measured. See eval-set.json for fixtures and pass rules.

Measurement and rollout

R1 - Handling effort and first-route quality

Business hypotheses, unknown baseline. Primary efficiency measure is active operator seconds spent reading/searching/editing/deciding, excluding unstaffed waits; divided by eligible reports, reported mean and p50/p95 per working week. Correct first-route fraction=adjudicated correct initial crew/category decisions / audited routable reports...

Why: Chosen to make the stated workflow explicit and bounded.

R2 - Safety, ownership and privacy guardrails

Count equipment/unauthorized/wrong-city actions and completion without durable receipt over all attempted actions, per day, with absolute zero-breach rule. Duplicate action fraction=report-revisions with >1 authoritative action / applied report-revisions. Urgent recall=adjudicated hazards escalated urgent / adjudicated hazards; acknowledgement time...

Why: Chosen to make the stated workflow explicit and bounded.

R3 - Shadow phase

Read approved deidentified/sandbox data and store recommendations locally; operators use existing process for actual work. No live C6 authority. Proposed two working weeks, extended for representation and powered design.

Why: Chosen to make the stated workflow explicit and bounded.

R4 - Assist display-only phase

Operators see recommendations/candidates but continue writes in current city UI. Proposed one working week usability and timing review after shadow gates. No new live ticket-writer integration until pilot.

Why: Chosen to make the stated workflow explicit and bounded.

R5 - Bounded pilot

Enable only operator-confirmed routine CREATE/LINK for one approved crew area; urgent/missing/conflicting reports remain manual. Two working weeks minimum and extend to computed power requirement; all five operators trained, eligibility strata logged.

Why: Chosen to make the stated workflow explicit and bounded.

R6 - Scale assisted scope

Expand areas sequentially after local policy/roster verification. End state remains operator-confirmed; no planned autonomous dispatch. Re-run EV20 for new policies/areas/languages.

Why: Chosen to make the stated workflow explicit and bounded.

R7 - Randomized effect evaluation

Compare recommendations visible versus hidden while both groups retain normal operator authority. Randomization unit is operator-shift cluster, balanced within operator and crew area over full weekly cycles; initial eligible routine reports only, excluding hazards from experiment but monitoring...

Why: Chosen to make the stated workflow explicit and bounded.

R8 - Authority, incident and residual-risk ownership

Proposed accountable roles, not invented appointees: service owner accepts launch/expansion; maintenance lead publishes routing/crew policy; supervisor owns manual/urgent queues; IT integration proves adapters; IT operations owns journals/kill switch/readback; security owns scopes/revocation; data owner approves retention; analyst owns measurement...

Why: Chosen to make the stated workflow explicit and bounded.

Build handoff and review

Start with build/00-overview.md. Load one milestone file at a time. Exact source rules remain in design.json; do not treat design approval as live write permission.

X1: Implement the proposed bounded workflow after city adapter/policy verification. This package implements no city system and grants no operational authority.

Review dimensions - evidence in review.json

Dimension	Score	Dimension	Score
D1	3/4	D7	3/4
D2	3/4	D8	3/4
D3	3/4	D9	3/4
D4	3/4	D10	3/4
D5	3/4	D11	3/4
D6	3/4	D12	3/4

ISSUE1 major resolved: W9/W10 terminal failure dispositions imply alternate states without matching next edges. Add explicit finalizer decision nodes and terminal receipt/cancellation nodes; bind C8 status guard...

ISSUE2 major resolved: Exact reasons required by EV3/EV11 and W8 are not fully declared as typed component outputs. Declare comprehensive reason producers and align negative fixtures with...

ISSUE3 major resolved: Additional variants in EV2/EV9/EV11/EV14/EV17 are prose assertions rather than fully resolved fixtures. Replace variant assertions with dedicated concrete goldens and include them in two-way...

ISSUE4 minor resolved: Current helper requires nonempty experiment on every rollout category, while the output contract permits inapplicable empty strings. Each card now contains a concrete R7...

Glossary

canonical: agreed

discriminated: typed

payload binding: approved details

digest: fingerprint

idempotency: duplicate-write protection

reciprocal: two-way

terminal disposition: final outcome

calc_digest: a fingerprint of the approved amount

After delivering DESIGN.pdf, ask the user: Would you also like a visual HTML version in the same ten-stage structure as the app? Generate HTML only after opt-in, unless already requested.

Full assumptions, owners and evidence: assumptions.md. Review method and failed rounds are preserved in review.json and review-round-*.json.